

NESL and NCB: Reference Manual

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Version 1.0.0.

Abstract

This document lists every setting recognised by `nesl`, the stagnation-line solver, and `nesl_condition_builder`, hereafter `ncb`, the parallel condition builder. Anything not listed here is ignored.

Command line

NESL

```
nesl [option]
```

<code>--help</code>	Displays the help message and exits.
<code>--job=<file></code>	Runs a simulation using a <code>.toml</code> configuration file.
<code>--grid=<file></code>	Builds the grid from a <code>.toml</code> configuration file and exits.

One option is given per run.

- `--job=<file>` runs the simulation described by the configuration file.
- `--grid=<file>` builds the grid from the same configuration file, writes `grid.png` and `grid.vtk`, and quits without solving the flow.
- Running with no argument, or with `--help`, prints the banner and exits.

NCB

```
nesl_condition_builder [options]
```

<code>--help</code>	Displays the help message and exits.
<code>--job=<file></code>	Template <code>.toml</code> configuration file.
<code>--bounds=<file></code>	Bounds to test, specified using a <code>.toml</code> configuration file.
<code>--max_cpus=<N></code>	Number of cores to spread the condition building across.

Arguments are read *by position*, not by name.

- `--job=<template.toml>` (first, required) must be a valid NESL job file.
- `--bounds=<ncb.toml>` (second, required) is the bounds file described under NCB bounds file settings.
- `--max_cpus=<N>` (third, optional) defaults to one process.

NESL job file settings

Required settings

The run aborts with a message if any of these is missing.

Key	Type	Description
GAS_MODEL	string	Path to the gas-model file.
MASS_FRAC	dict	Freestream mass fractions, keyed by species name.
T_INF	float	Freestream temperature, K.
U_INF	float	Freestream velocity, m/s.
RADIUS	float	Body nose radius, m.
DOMAIN	float	Length of the stagnation-line domain, m.
CELLS	int	Number of cells in the grid.

In addition, exactly one of `P_INF` or `RHO_INF` must be supplied; the run aborts if neither is present.

Key	Type	Default	Description
P_INF	float	None	Freestream pressure, Pa.
RHO_INF	float	None	Freestream density, kg/m ³ .

If both are given, `P_INF` is used.

Species-keyed dictionaries

`MASS_FRAC`, `WALL_MASS_FRAC` and `RECOM_RATE` are all dictionaries keyed by species name. Names must match the gas model exactly, and *any species absent from the dictionary is set to zero*.

```
MASS_FRAC = {N2 = 0.767, O2 = 0.233}
```

Thermochemistry

Key	Type	Default	Description
CHEM_MODEL	string	<i>absent</i>	Path to the reaction file. Chemistry is on when this is given and off when it is not.
EXCHANGE_MODEL	string	" "	Path to the energy-exchange file, for two-temperature runs.
T_FROZEN	float	1.0	Cells below this temperature are frozen and no chemistry occurs.

Geometry and wall

Key	Type / values	Default	Description
AXISYMMETRIC	bool	true	true for a sphere, false for a cylinder.
CURVATURE	string	"empirical"	Method to determine radius of curvature of the shock.
T_WALL	float	-1.0	Isothermal wall temperature, K. Only takes effect with <code>VISCOSITY</code> on; if <code>VISCOSITY</code> is on and <code>T_WALL</code> is not set, the wall is adiabatic.

Curvature models

- **"empirical"** Uses an empirical correlation to determine, R_s .
- **"thin_shock_layer"** Relies on the simple approximation, $R_s = R_b + \Delta$. Can under-predict shock standoff for lower density ratios.

Viscous and wall-catalysis settings

Key	Type / values	Default	Description
VISCOSITY	bool	false	Turns on viscous effects.
SENSOR	bool	true	Jameson pressure sensor that damps the viscous fluxes near the shock. Improves stability on startup, but valid for steady runs only.
BETA_SMOOTHING	bool	true	Applies smoothing to β across the shock to account for a finite shock thickness.
LEWIS_NUM	float	1.0	Lewis number used for species diffusion.
CATALYTIC	string	"non_catalytic"	Wall catalysis model.
WALL_MASS_FRAC	dict	empty	Wall mass fractions, keyed by species. Used only by "fixed_composition".
RECOM_RATE	dict	absent	Per-species recombination data. Used only by "finite_rate".

Catalysis models

- "non_catalytic" Fixes a zero species gradient across the wall.
- "finite_rate" Each species recombines at the wall at a rate set by its efficiency, defined in RECOM_RATE.
- "fixed_composition" The wall composition is held at WALL_MASS_FRAC.
- "equilibrium" The wall composition is held at the equilibrium composition for the stagnation pressure and wall temperature.

RECOM_RATE format

Each catalytic species maps to a dictionary carrying an efficiency and the species the recombined mass is deposited into.

```
RECOM_RATE = {O = {efficiency = 0.01, product = "O2"}, N = {efficiency = 0.005, product = "N2"}}
```

- **efficiency** (float) is the recombination efficiency used in calculating the catalytic velocity.
- **product** is a species name that must exist in the gas model. The mass consumed by the species is added to this species.
- Species omitted from the dictionary get an efficiency of zero, so they do not recombine.

Grid

Key	Type / values	Default	Description
CLUSTERING	false, int, or float	false	A number greater than 1 is the Roberts clustering parameter, clustering cells towards the wall; values close to 1 cluster hardest.

There are only two accepted values.

- **false**, or the key left out altogether, builds a uniform grid; fixed Δx .
- **A number greater than 1** clusters cells towards the wall. Values close to 1 cluster hardest.

Time stepping and termination

Key	Type / values	Default	Description
T_END	float, or dict {steady = <residual>}	{steady = 1e-1}	A float runs time-accurately and stops at that simulation time. The dictionary form runs in steady mode, stopping once the largest change in the solution has stayed below that residual for 100 iterations.
MAX_ITERS	int	1e8	Maximum number of iterations; the run stops and reports when it is reached.
CFL_c	float	0.4	Convective CFL number, $\Delta t_c = \text{CFL}_c \Delta x / (u + a)$.
CFL_d	float	0.4	Diffusive CFL number, $\Delta t_d = \text{CFL}_d \Delta x^2 / \max(\nu, \alpha, D_s)$.
LTS	bool	false	Local time stepping. Each cell advances on its own stable time step, which accelerates convergence but destroys time accuracy.
HOT_START	bool	true	Starts the run from an estimate of the post-shock state instead of a uniform freestream, which converges faster.

LTS has two requirements, and the run stops if either is broken.

- It needs steady mode, so T_END must be given in its dictionary form; and
- it cannot be combined with transient output, that is, the dictionary form of SAVE.

Output

Key	Type / values	Default	Description
SAVE	false, a string, or a dict {path = <string>, interval = <float>}	false	Output control, described below.
PLOT	bool	false	Writes PNG plots after the run. Has no effect unless SAVE is set.
VERBOSE	bool	true	Prints progress, the iteration and residual trace every 1000 iterations, and the final heat flux and standoff.

SAVE behaves in three ways.

- **false**, nothing is written.
- **A string**, the profile is written to that path as a CSV, and the stagnation-point summary.
- **A dictionary**, as above, plus a transient trace of the same stagnation-point quantities, written to `transient_<path>` at the given `interval` of simulation time. In this mode the `stagpoint_` file is *not* written, which matters when the run is driven by NCB (see Output layout).

The profile CSV columns are $x, u, \rho, T_{tr}, p, e, \mu, k_{tr}, a, \beta, T_{ve}, k_{ve}$, followed by one column per species. On a one-temperature gas model the two vibro-electronic columns are omitted and the header reads T and k .

With PLOT enabled, the plots written are `<name>_prim.png` for the primitives, `<name>_massf.png` if chemistry is on, and `<name>_transient.png` if the transient mode is in use, where `<name>` is SAVE with its extension stripped.

A complete NESL job file

```
# Freestream.
T_INF = 300.0
P_INF = 100.0
U_INF = 5000.0
MASS_FRAC = {N2 = 0.767, O2 = 0.233}

# Geometry.
RADIUS = 0.05
DOMAIN = 0.005
AXISYMMETRIC = true

# Models.
GAS_MODEL = "air-5sp.gas"
CHEM_MODEL = "air-5sp.chem"
EXCHANGE_MODEL = "air-5sp.kin"

# Grid.
CELLS = 100
CLUSTERING = 1.05

# Wall.
T_WALL = 300.0
VISCOSITY = true
CATALYTIC = "equilibrium"
LEWIS_NUM = 1.2

# Time stepping.
T_END = {steady = 1e-2}
MAX_ITERS = 1e7
CFL_c = 0.4
CFL_d = 0.1
LTS = true

# Output.
SAVE = "result.csv"
PLOT = true
VERBOSE = true
```

Invoked as,

```
nesl --job=template.toml
```

NCB bounds file settings

The bounds file defines the sweep. NCB takes the cartesian product, or the element-wise zip, of the freestream arrays, generates one NESL job per combination, runs them in parallel, and collects the results.

NCB is not configured to work with transient runs. It only handles the steady stagnation-point output, so the template must use the plain string form of **SAVE**. See Output layout.

Keys

Key	Type	Required	Default	Description
T_BOUND	dict, string, or number	Yes	-	Freestream temperature values, K.

Key	Type	Required	Default	Description
U_BOUND	dict, string, or number	Yes	-	Freestream velocity values, m/s.
P_BOUND	dict, string, or number	One of the two	-	Freestream pressure values, Pa.
RHO_BOUND	dict, string, or number	One of the two	-	Freestream density values, kg/m ³ .
PERMUTATIONS	bool	No	true	true sweeps the full cartesian product of the three arrays. false zips them element-wise, so entry <i>i</i> of each array forms case <i>i</i> .
DOMAIN_SCALE	float	No	1.5	Safety factor on the estimated shock standoff used to set each case's DOMAIN. Ignored when FIXED_DOMAIN is set.
FIXED_DOMAIN	bool	No	false	true holds the domain fixed across the sweep: NCB estimates no standoff and leaves the template's DOMAIN untouched.

One of P_BOUND or RHO_BOUND must be given. If P_BOUND is present it is used and the jobs are written with P_INF; otherwise RHO_BOUND is used and the jobs are written with RHO_INF.

Under PERMUTATIONS = **false** the arrays should be the same length; any extra entries are ignored.

Bound specification

Each of T_BOUND, U_BOUND, P_BOUND and RHO_BOUND accepts three forms.

A single number, a scalar that is held fixed across the sweep.

```
T_BOUND = 300.0
```

A range dictionary, takes lower, upper, cells and bound_type.

```
U_BOUND = {lower = 4000.0, upper = 5000.0, cells = 5, bound_type = 'linear'}
```

bound_type	Spacing
'linear'	cells points evenly spaced from lower to upper.
'log_lower'	cells points logarithmically spaced, ascending from lower to upper, so points bunch near lower.
'log_upper'	cells points logarithmically spaced, descending from upper to lower, so points bunch near upper.

A path string, a text file of values, one per line, loaded directly.

```
T_BOUND = "temperatures.txt"
```

What NCB writes into each job

For each case NCB copies the template and overrides:

- **T_INF**, **U_INF**, and either **P_INF** or **RHO_INF** from the sweep;
- **DOMAIN**, computed per case as **DOMAIN_SCALE** times the inviscid frozen normal-shock standoff estimate for that condition. Under **FIXED_DOMAIN = true** this key is not written at all and the template's own **DOMAIN** carries through to every case; and
- **GAS_MODEL**, plus **CHEM_MODEL** and **EXCHANGE_MODEL** where present, each rewritten to `../../<file name>`.

That last rewrite means the model files must sit in the directory NCB is launched from, since each case runs two levels down in `runs/caseNNN/`.

Everything else in the template - grid, wall, catalysis, time stepping, output - is passed through unchanged.

Output layout

```
runs/
  case00/
    job.toml
    ... nesl output ...
  case01/
    results.csv
```

NCB refuses to start if `runs/` already exists.

`results.csv` collects one row per case: case number, `T_inf`, `p_inf` or `rho_inf`, `u_inf`, standoff, convective heat flux and diffusive heat flux.

The template's **SAVE** must be a plain string. With `false`, or with the dictionary form that enables the transient trace, every row of `results.csv` comes out as NaN.

A complete NCB bounds file

```
T_BOUND = {lower = 200.0, upper = 300.0, cells = 3, bound_type = 'linear'}
RHO_BOUND = {lower = 1e-4, upper = 1e-2, cells = 4, bound_type = 'log_lower'}
U_BOUND = {lower = 4000.0, upper = 8000.0, cells = 5, bound_type = 'log_upper'}
PERMUTATIONS = true
DOMAIN_SCALE = 1.5
```

Invoked as,

```
nesl_condition_builder --job=template.toml --bounds=bounds.toml --max_cpus=8
```

Improving stability

Settings worth adjusting first when a run will not converge.

- Lower **CFL_c** and **CFL_d**, e.g. to 0.1-0.2.
- Keep **SENSOR** and **HOT_START** on.
- Cluster the grid towards the wall with **CLUSTERING**, close to 1, e.g. 1.01-1.05.
- Increase **CELLS** to better resolve the shock and boundary layer.
- Check **DOMAIN** is long enough to contain the shock.